



— LAUGHTALE · CCNA 200-301 V1.1 · PRINTABLE PACK

Blueprint *Breakdown*

All 53 exam sub-objectives with verb class, what to master, depth limits, high-yield ranking and the do-not-study list.

53

Sub-objectives

13

High-yield

6

Domains

DOMAIN 1 · FUNDAMENTALS

DOMAIN 2 · ACCESS

DOMAIN 3 · IP CONNECTIVITY

DOMAIN 4-6 · SERVICES / SECURITY / AUTOMATION

★ HIGH YIELD marks a disproportionate point share — over-invest there

STOP blocks are hard depth limits, not suggestions

PRINT · A4 PORTRAIT · SINGLE COLUMN · BACKGROUND GRAPHICS ON

Overview

Part of the CCNA 200-301 v1.1 Study Master Plan — see 00_Index_and_How_To_Use.md for notation/checkbox conventions.
Next: 02_Study_Calendar.md.

Section 1 — Exam Blueprint Breakdown

Domain 1.0 — Network Fundamentals (20%)

Thirteen objectives. Two of them (1.6, 1.8) carry more weight than the other eleven combined, because subnetting math bleeds into every other domain on the exam.

1.1 Explain the role and function of network components

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 1

- Which OSI layer each device makes its forwarding decision at: router (L3/IP), L2 switch (L2/MAC), L3 switch (both, in hardware), NGFW (up to L7), IPS (inline, L7 inspection, blocks in real time), AP (802.11 ↔ 802.3 conversion), WLC (centralised control of lightweight APs via CAPWAP), endpoint, server.
- PoE standards and budgets: 802.3af (15.4 W), 802.3at / PoE+ (30 W), 802.3bt / UPoE and UPoE+ (60 W / 90 W). Know power is delivered over the same twisted pair as data, and typical draws (IP phone, AP, camera).
- IDS vs IPS: IDS is out-of-band and alerts; IPS is inline and drops.

STOP • DEPTH LIMIT

Do not learn specific Cisco model numbers, Firepower policy configuration, or IPS signature tuning. Do not configure PoE — there is no configure verb here.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.2 Describe characteristics of network topology architectures

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 52 (LAN ARCHITECTURES), DAY 53 (WAN ARCHITECTURES)

- Two-tier (collapsed core): access + collapsed distribution/core. When it's the right answer — small to medium campus, cost-constrained.
- Three-tier: access (endpoint connectivity, PoE, port security), distribution (routing, VLAN boundary, policy, L2/L3 demarcation), core (high-speed transport, no policy, no filtering).
- Spine-leaf: every leaf connects to every spine, never leaf-to-leaf or spine-to-spine. Two hops maximum between any endpoints = predictable latency. Solves east-west traffic and removes STP-blocked-link waste. Data centre answer.

- WAN: leased line, MPLS L2VPN/L3VPN, Metro Ethernet, internet VPN, SD-WAN. Know what each is at a one-paragraph level.
- SOHO: single integrated box — router + switch + AP + firewall + NAT.
- On-premises vs cloud: IaaS / PaaS / SaaS responsibility boundaries, and the trade-offs (CapEx vs OpEx, control vs elasticity).

STOP • DEPTH LIMIT

No ACI, no VXLAN encapsulation detail, no MPLS label mechanics, no SD-WAN vEdge/vSmart component naming, no AWS/Azure service names.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.3 Compare physical interface and cabling types

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 2

- Single-mode fibre: small core (~9 µm), laser source, long distance (km range), more expensive. Multimode: larger core (50/62.5 µm), LED/VCSEL, shorter distance, cheaper.
- Copper twisted pair: Cat5e (1 Gbps), Cat6 (10 Gbps to 55 m), Cat6a (10 Gbps to 100 m). 100 m limit for all.
- Straight-through vs crossover and which device pairs need which (then know that MDIX auto makes it mostly moot on modern gear).
- Shared media (hub, half-duplex, one collision domain) vs point-to-point (switch, full-duplex, no collisions).

STOP • DEPTH LIMIT

Do not memorise every IEEE standard name (1000BASE-LX10 vs -EX vs -ZX distance tables). Know the categories and the reasoning, not the catalogue.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.4 Identify interface and cable issues (collisions, errors, duplex/speed mismatch)

CLASS [INTERPRET]

YIELD MEDIUM

JEREMY DAY 9 (SWITCH INTERFACES)

- Read show interfaces output and name the fault from the counters. This is the actual exam skill.
- CRC errors → damaged cable, EMI, bad transceiver. CRC with no collisions often means duplex mismatch on the receiving side.
- Runts (<64 bytes) → collisions or duplex mismatch. Giants (>1518) → MTU/oversized frame issue.
- Late collisions → duplex mismatch (classic) or cable run exceeding spec.
- Input errors vs output errors vs drops — which direction is broken.
- Speed mismatch = link does not come up at all. Duplex mismatch = link comes up and performs terribly. Know this distinction cold; it's a favourite question.
- show interfaces status for a fast up/down/err-disabled and speed/duplex sweep.

STOP • DEPTH LIMIT

No optical power budget calculations, no TDR cable testing output, no SFP diagnostics.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.5 Compare TCP to UDP

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 30

- TCP: connection-oriented, three-way handshake (SYN, SYN-ACK, ACK), four-way termination (FIN/ACK ×2), sequence and acknowledgement numbers, sliding window / flow control, retransmission, 20-byte minimum header.

- UDP: connectionless, no handshake, no ordering, no retransmission, 8-byte header, four fields only (source port, dest port, length, checksum).
- Header fields for both, in order. You will get asked what a specific field does.
- Port numbers you must know by heart: FTP 20/21, SSH 22, Telnet 23, SMTP 25, DNS 53, DHCP 67/68, TFTP 69, HTTP 80, POP3 110, NTP 123, SNMP 161/162, HTTPS 443, Syslog 514, RADIUS 1812/1813, TACACS+ 49.
- Which applications pick which and why (VoIP/video/DNS/DHCP/TFTP → UDP; web/mail/file transfer/SSH → TCP).

STOP • DEPTH LIMIT

No congestion control algorithm detail (Reno, CUBIC, slow start maths). No TCP option fields beyond MSS awareness.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.6 Configure and verify IPv4 addressing and subnetting ★ HIGH YIELD

CLASS [CONFIG]

YIELD HIGHEST IN THE EXAM

JEREMY DAYS 7, 8, 13, 14, 15

- Given any address and prefix, produce in ≤45 seconds: network ID, broadcast address, first usable host, last usable host, usable host count, dotted-decimal mask, and the next subnet's network ID.
- The reverse direction: given a host requirement, pick the correct prefix. Given a subnet count requirement, pick the correct prefix.
- VLSM: carve one block into differently-sized subnets in descending size order without overlap.
- Route summarisation: given a set of networks, produce the summary prefix.
- CLI: address a router interface, address an SVI, address a subinterface, no shutdown, and verify with show ip interface brief, show ip interface, show ip route, show running-config interface.
- Recognise up/up, up/down, administratively down/down and what each implies about layer 1 vs layer 2 vs configuration.

STOP • DEPTH LIMIT

Nothing. There is no rabbit hole here — this is the one objective where more time is always correct. It also underpins ACLs, OSPF wildcard masks, NAT pools, and DHCP scopes.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.7 Describe private IPv4 addressing

CLASS [CONCEPT]

YIELD LOW (BUT ASSUMED EVERYWHERE)

JEREMY DAY 7

- RFC 1918: 10.0.0.0/8, 172.16.0.0/12 (172.16.0.0 – 172.31.255.255 — know the /12 boundary, it's a trap), 192.168.0.0/16.
- Not routable on the public internet; requires NAT to reach it.
- Adjacent special ranges you should recognise on sight: 127.0.0.0/8 loopback, 169.254.0.0/16 APIPA/link-local (what it means when a client has one — DHCP failed), 224.0.0.0/4 multicast.

STOP • DEPTH LIMIT

No RFC 6598 CGNAT (100.64.0.0/10) depth, no full IANA special-purpose registry.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.8 Configure and verify IPv6 addressing and prefix ★ HIGH YIELD

CLASS [CONFIG]

YIELD HIGH

JEREMY DAYS 31, 32, 33

- Address compression and expansion rules, in both directions, flawlessly. Leading-zero removal, single :: only, and the "longest run of zeros" tie-break.
- IPv6 subnetting on nibble boundaries (/48 → /52 → /56 → /60 → /64). Know why /64 is the standard host prefix.

- CLI: enable IPv6 routing globally, assign a static global unicast address, assign an address using EUI-64, assign a link-local address manually, configure an IPv6 static route and default route.
- Verify with show ipv6 interface brief, show ipv6 interface, show ipv6 route, and IPv6 ping including pinging a link-local address (which requires specifying the exit interface).
- Understand that an interface has multiple IPv6 addresses simultaneously (link-local + one or more global) and which is used as source for what.

STOP • DEPTH LIMIT

No DHCPv6 server configuration, no SLAAC deep mechanics beyond RA/RS purpose, no OSPFv3, no IPv6 ACLs, no NAT64/DS-Lite/tunnelling.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.9 Describe IPv6 address types

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAYS 31, 32

- Identify type from the leading bits on sight: GUA 2000::/3, ULA fc00::/7 (practically fd00::/8), link-local fe80::/10, multicast ff00::/8, unspecified ::, loopback ::1.
- Anycast: same address on multiple nodes, routed to the topologically nearest. Note there is no IPv6 broadcast — multicast replaces it.
- Well-known multicast groups: ff02::1 all nodes, ff02::2 all routers, ff02::5/ff02::6 OSPF, solicited-node ff02::1:ff00:0/104.
- Modified EUI-64, performed by hand: split the 48-bit MAC, insert FFFE in the middle, flip the 7th bit of the first byte (the U/L bit). Be able to do this in both directions.

STOP • DEPTH LIMIT

No multicast routing, no MLD, no Neighbour Discovery packet-by-packet breakdown beyond NS/NA/RS/RA purpose.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.10 Verify IP parameters for Client OS (Windows, macOS, Linux)

CLASS [CONFIG] (VERIFY ONLY)

YIELD LOW-MEDIUM

JEREMY DAY 32 / LAB VIDEOS

- Windows: ipconfig /all (IP, mask, gateway, DNS, MAC, DHCP server), ping, tracert, nslookup, arp -a.
- Linux: ip addr, ip route, ip neigh, dig, legacy ifconfig.
- macOS: ifconfig, networksetup -getinfo <service>.
- The actual exam skill: read client output and spot the misconfiguration — wrong gateway (not in the client's own subnet), wrong mask (breaks local vs remote decisions), missing DNS, or a 169.254.x.x address meaning DHCP never answered.

STOP • DEPTH LIMIT

No OS-specific network stack tuning, no netsh scripting, no systemd-networkd or NetworkManager config files.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.11 Describe wireless principles

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 55

- Non-overlapping channels: 2.4 GHz has only 1, 6, 11 at 20 MHz. 5 GHz has many (24+ depending on DFS/regulatory domain). Be able to explain co-channel interference vs adjacent-channel interference and which is worse.
- SSID vs BSSID vs ESS: SSID is the name, BSSID is the AP radio's MAC, ESS is multiple APs sharing an SSID.

- RF behaviour: attenuation, absorption, reflection, refraction, scattering, multipath; SNR and why it matters more than raw signal strength; the relationship between distance, data rate, and modulation.
- Encryption/authentication: WPA2-Personal (PSK) vs WPA2-Enterprise (802.1X/RADIUS); AES-CCMP; WPA3 with SAE and its improvement over the WPA2 four-way handshake.
- 2.4 GHz vs 5 GHz trade-off: range/penetration vs capacity/interference.

STOP • DEPTH LIMIT

No 802.11 frame-type breakdown, no antenna gain dBi maths, no RF site-survey methodology, no specific 802.11ax/be feature tables.

☐ STUDIED ☐ ANKI ☐ PROVEN

1.12 Explain virtualization fundamentals (server virtualization, containers, VRFs)

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 54 (PARTS 1-3)

- Type 1 hypervisor (bare metal — ESXi, Hyper-V, KVM) vs Type 2 (hosted — VirtualBox, VMware Workstation). Which is used in a data centre and why.
- Virtual switch role: how VMs reach the physical network, and why the physical uplink is typically a trunk.
- Containers vs VMs: containers share the host kernel, no guest OS, seconds to start, megabytes not gigabytes. VMs give stronger isolation.
- VRF: multiple independent routing tables on one device. The one-line framing that wins questions — "VRFs are to Layer 3 what VLANs are to Layer 2."

STOP • DEPTH LIMIT

No vSphere/ESXi administration, no Docker commands, no Kubernetes at all, no VRF-lite configuration (there is no configure verb).

☐ STUDIED ☐ ANKI ☐ PROVEN

1.13 Describe switching concepts

CLASS [CONCEPT]

YIELD MEDIUM-HIGH

JEREMY DAYS 5, 6

- MAC learning: the switch reads the source MAC of an incoming frame and maps it to the ingress port and VLAN. Aging default 300 seconds on Cisco.
- Frame switching: destination MAC found in the CAM table → forward out that one port only.
- Frame flooding: destination is broadcast, unknown unicast, or multicast → out every port in the VLAN except the ingress port.
- MAC address table / CAM table structure: MAC ↔ port ↔ VLAN ↔ type (dynamic/static).
- The exam skill: given a topology and a specific sequence of frames, state the exact contents of each switch's MAC table afterwards, and state for each frame whether it was switched or flooded. Practise this on paper — it is a guaranteed question type.
- show mac address-table, and clear mac address-table dynamic.

STOP • DEPTH LIMIT

No store-and-forward vs cut-through vs fragment-free switching detail (removed from current scope), no TCAM/ASIC architecture.

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 2.0 — Network Access (20%)

The heaviest simulation domain. VLANs, trunking, EtherChannel, and STP interpretation are where performance-based items concentrate. Note that in v1.1, objective 2.9 was softened from configure to interpret — the wireless GUI is now a reading exercise, not a building one.

2.1 Configure and verify VLANs (normal range) spanning multiple switches ★ HIGH YIELD

CLASS [CONFIG] YIELD VERY HIGH JEREMY DAYS 16, 17, 18

- Create a VLAN, name it, assign an access port, verify with show vlan brief.
- Data vs voice access ports on the same physical interface: a voice VLAN carries tagged voice traffic while the data VLAN stays untagged on the same port.
- Default VLAN 1 behaviour, why it's poor practice to use it, and normal range (1–1005) vs extended range (1006–4094 — awareness only).
- VLANs spanning multiple switches: the same VLAN ID must exist on every switch in the path and be allowed on the trunks between them. A missing VLAN on a transit switch is the single most common lab failure — expect to troubleshoot exactly this.
- InterVLAN connectivity: router-on-a-stick (subinterfaces with encapsulation dot1q) and L3 switch SVIs (ip routing + SVI per VLAN). Know when each is appropriate.
- Verify with show vlan brief, show interfaces switchport, show interfaces trunk, show ip route.

STOP • DEPTH LIMIT

No VTP configuration (Jeremy covers it in Day 19 for context — watch once, do not lab it; it is not on the v1.1 blueprint). No private VLANs. No extended-range VLAN configuration.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.2 Configure and verify interswitch connectivity ★ HIGH YIELD

CLASS [CONFIG] YIELD VERY HIGH JEREMY DAYS 17, 18

- Statically configure a trunk port (mode trunk, and where required set the encapsulation first on older platforms).
- 802.1Q tag structure: 4 bytes inserted after the source MAC, 12-bit VLAN ID field, TPID 0x8100. Native VLAN frames leave untagged.
- Native VLAN: change it, and understand the security reason (VLAN hopping / double-tagging) and the operational reason (a native VLAN mismatch generates errors and merges two VLANs' traffic).
- Prune the allowed VLAN list on a trunk, and add/remove from it correctly without wiping the list.
- Troubleshoot a trunk that won't form: mode mismatch, native VLAN mismatch, VLAN not allowed, VLAN not defined.
- Verify with show interfaces trunk (the single most useful switching command on the exam) and show interfaces switchport.

STOP • DEPTH LIMIT

DTP exists and you should know what dynamic auto/dynamic desirable do and why hardcoding is better practice — but do not build a lab around DTP negotiation states. No ISL (dead).

☐ STUDIED ☐ ANKI ☐ PROVEN

2.3 Configure and verify Layer 2 discovery protocols (CDP and LLDP)

- Enable/disable CDP globally and per-interface. Same for LLDP — and note LLDP is disabled by default on Cisco while CDP is enabled by default.
- LLDP's transmit and receive are configured separately per interface. This is a favourite exam detail.
- Default timers: CDP 60 s hello / 180 s hold. LLDP 30 s / 120 s.
- Read show cdp neighbors and show cdp neighbors detail (the detail version gives the neighbour's IP address — that's why it matters). Same for show lldp neighbors / detail.
- Use the output to reconstruct a topology diagram. That is the exam task: "given this CDP output, which device is connected to which port."
- Security angle: CDP leaks device model and IOS version, so disable it on untrusted-facing ports.

STOP • DEPTH LIMIT

No CDP/LLDP TLV field enumeration.

☐ STUDIED ☐ ANKI ☐ PROVEN
2.4 Configure and verify (Layer 2/Layer 3) EtherChannel (LACP)

★ HIGH YIELD

- LACP modes: active and passive. Active/active and active/passive form a channel; passive/passive does not. PAGP (desirable/auto) and static on exist — know the mode-combination matrix.
- Layer 2 EtherChannel: bundle physical ports into a port-channel, then configure trunking/VLAN settings on the port-channel interface, not the members.
- Layer 3 EtherChannel: no switchport on members, create the port-channel, apply the IP address to the port-channel interface.
- The consistency requirement: every member port must match on speed, duplex, mode (access/trunk), native VLAN, and allowed VLAN list. Mismatch = port excluded from the bundle. This is the standard troubleshooting scenario.
- Load-balancing methods (src-mac, dst-mac, src-dst-ip etc.) — know that the hash determines which physical link a flow uses, and that a single flow never spans links.
- Verify with show etherchannel summary (read the flags — P bundled, D down, s suspended, I stand-alone), show etherchannel port-channel, show interfaces port-channel.

STOP • DEPTH LIMIT

No advanced hashing algorithm internals, no cross-stack/MLAG/vPC.

☐ STUDIED ☐ ANKI ☐ PROVEN
2.5 Interpret basic operations of Rapid PVST+ Spanning Tree Protocol

★ HIGH YIELD

- Root bridge election: lowest bridge ID wins = priority (default 32768) + extended system ID (the VLAN number) + MAC address. Compute this by hand from a diagram. Priority must be set in increments of 4096.
- Root port selection order: lowest root path cost → lowest sender bridge ID → lowest sender port ID → lowest local port ID. Know the tie-break order in sequence.
- Port cost defaults for RSTP/short mode: 10 Mbps = 100, 100 Mbps = 19, 1 Gbps = 4, 10 Gbps = 2.
- Port roles: root, designated, alternate, backup, disabled. Port states in RSTP: discarding, learning, forwarding (RSTP collapses blocking/listening into discarding — be able to map the legacy STP states to the RSTP ones).
- Link types: point-to-point (full duplex), shared (half duplex), edge (PortFast).
- PortFast: skip straight to forwarding on an access port with an endpoint. Never on a switch-to-switch link.
- Protection features and what each stops: BPDU guard (err-disables a PortFast port that receives a BPDU), BPDU filter (suppresses BPDUs), root guard (blocks a superior BPDU from taking over as root, puts port in root-inconsistent), loop guard (protects against unidirectional link failure causing a blocked port to wrongly go forwarding).
- Set primary and secondary root for a VLAN and know what that command actually does to the priority value.

- Verify with show spanning-tree, show spanning-tree vlan <id>, show spanning-tree root, show spanning-tree interface <int> detail.
- The exam skill: given a diagram with bridge priorities and MACs, mark every port's role and state. Do this by hand, repeatedly.

STOP • DEPTH LIMIT

The verb is interpret, not configure. You still need the root-priority and PortFast/BPDU-guard commands, but do not study MST, do not study PVST+ legacy timer tuning (hello/forward delay/max age maths), and do not study STP convergence timer optimisation.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.6 Describe Cisco Wireless Architectures and AP modes

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 56

- Autonomous AP (standalone, self-configured) vs lightweight AP + WLC (split-MAC, centralised).
- Split-MAC: which functions stay on the AP (real-time — beacons, ACKs, encryption, buffering) and which move to the WLC (management — association, authentication, RRM, security policy).
- CAPWAP: two tunnels, control (UDP 5246) and data (UDP 5247). Tunnels the traffic from AP to WLC.
- WLC deployment models: unified/on-prem appliance, embedded, cloud-based, Mobility Express, and the concept of a centralised vs FlexConnect data path.
- AP modes: local (default), FlexConnect, monitor, sniffer, rogue detector, SE-Connect, bridge/mesh. Know which one you'd pick for a given requirement (e.g. a branch site with a slow WAN link → FlexConnect).

STOP • DEPTH LIMIT

No WLC HA/SSO configuration, no RRM algorithm internals, no mesh RAP/MAP design.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.7 Describe physical infrastructure connections of WLAN components

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 56

- AP switchport: access port if one SSID/VLAN and traffic is tunnelled to the WLC in local mode; trunk port when the AP is autonomous or in FlexConnect with local switching of multiple VLANs. Be able to justify which and why.
- WLC ports: distribution system port, service port, redundancy port, console. WLC interfaces: management, virtual, dynamic (per-WLAN), AP-manager.
- LAG on the WLC: bundles distribution ports into one logical link; know that it's typically all-or-nothing and requires a matching EtherChannel on the switch side.
- PoE feeding the AP from the access switch, and the power-budget consequence.

STOP • DEPTH LIMIT

No WLC port-by-port cabling specifications per hardware model.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.8 Describe network device management access

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 4, DAY 42, DAY 48

- Console (out-of-band, physical, works with no network), Telnet (TCP 23, cleartext — never use), SSH (TCP 22, encrypted — always use), HTTP (80) vs HTTPS (443) for GUI access.

- TACACS+ vs RADIUS: TACACS+ is Cisco-originated, TCP 49, encrypts the entire payload, and separates authentication/authorisation/accounting — so it's the device-administration choice. RADIUS is UDP 1812/1813, encrypts only the password, and combines authentication and authorisation — so it's the network-access/802.1X choice.
- Cloud-managed (Meraki-style): device phones home to a cloud controller; separation of management plane from data plane.
- In-band vs out-of-band management, and why OOB matters when the network is broken.

STOP • DEPTH LIMIT

No ISE policy configuration, no TACACS+ packet format.

☐ STUDIED ☐ ANKI ☐ PROVEN

2.9 Interpret the wireless LAN GUI configuration for client connectivity

CLASS [INTERPRET]

YIELD MEDIUM

JEREMY DAY 58

- Walk the WLC GUI WLAN creation flow and be able to read a screenshot and say what it will do: WLAN ID/SSID/profile name, the interface (which maps the WLAN to a VLAN), status enable/disable, broadcast SSID on/off.
- Security tab: Layer 2 security choice (WPA2/WPA3, PSK vs 802.1X), AKM, encryption cipher, the PSK field, and where a RADIUS server gets attached for Enterprise.
- QoS tab: Platinum (voice) / Gold (video) / Silver (best effort, default) / Bronze (background). Know the ordering.
- Advanced tab: session timeout, client exclusion, DHCP required, FlexConnect local switching, band select, client load balancing.

STOP • DEPTH LIMIT

The v1.1 verb is interpret, not configure. You are expected to read and reason about GUI screens, not build them from memory. Do not memorise menu paths for every WLC firmware version. Objective 5.10 is the one place a GUI configure verb still applies (WPA2 PSK).

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 3.0 — IP Connectivity (25%)

The largest single domain. Only five objectives, but three of them (3.1, 3.3, 3.4) are dense and heavily simulated. OSPFv2 is the highest-value single technology on the exam.

3.1 Interpret the components of routing table ★ HIGH YIELD

CLASS [INTERPRET] YIELD VERY HIGH JEREMY DAY 11 (ROUTING FUNDAMENTALS)

- Read any line of show ip route and name every field. Given O 192.168.10.0/24 [110/65] via 10.0.0.2, 00:04:11, GigabitEthernet0/1, you must instantly identify: source code, prefix, mask, administrative distance (110), metric (65), next hop, uptime, exit interface.
- Routing protocol codes: C connected, L local, S static, S* static default, O OSPF, O IA OSPF inter-area, D EIGRP, R RIP, B BGP.
- Default administrative distances, memorised: connected 0, static 1, eBGP 20, EIGRP 90, OSPF 110, RIP 120, external EIGRP 170, iBGP 200, unreachable 255.
- Gateway of last resort: what sets it, how it displays, and what happens when it isn't set.
- The difference between a C connected route and its paired L local /32 host route.
- show ip route <prefix> for the detailed single-route view.

STOP • DEPTH LIMIT

No BGP path-attribute detail, no EIGRP metric formula (K-values), no route redistribution.

☐ STUDIED ☐ ANKI ☐ PROVEN

3.2 Determine how a router makes a forwarding decision by default ★ HIGH YIELD

CLASS [INTERPRET] YIELD HIGH JEREMY DAYS 11, 12 (LIFE OF A PACKET)

- The decision order, in this exact sequence: 1. Longest prefix match (most specific route wins, full stop — AD and metric never override a longer match). 2. Administrative distance (used only to choose between routes to the same prefix learned from different sources). 3. Metric (used only to choose between routes to the same prefix from the same protocol).
- Given a routing table and a destination IP, state the exact chosen route. This appears in some form on almost every attempt.
- Equal-cost multi-path: same prefix, same AD, same metric → both installed, load balanced.
- Life of a packet end to end: ARP, MAC rewrite at every hop, TTL decrement, source/destination IP never changing (absent NAT). Being able to narrate this is worth a surprising number of points.

STOP • DEPTH LIMIT

No CEF/FIB/adjacency-table internals, no per-packet vs per-destination load-balancing platform behaviour.

☐ STUDIED ☐ ANKI ☐ PROVEN

3.3 Configure and verify IPv4 and IPv6 static routing ★ HIGH YIELD

CLASS [CONFIG] YIELD VERY HIGH JEREMY DAY 11 (STATIC ROUTING), DAY 33

- Network route (a subnet), host route (/32 or /128), default route (0.0.0.0/0 and ::/0).
- Three ways to specify a static route — next-hop IP only, exit interface only, or both. Know the recursive-lookup cost of next-hop-only, and why exit-interface-only on a multi-access (Ethernet) segment causes ARP problems. Fully-specified is safest on Ethernet.

- Floating static: raise the AD above the dynamic protocol's so it only installs when the primary route dies. This is the standard backup-link question.
- IPv6 equivalents including a static route via a link-local next hop (which requires the exit interface to be specified).
- Verify: show ip route static, show ipv6 route static, and prove it by shutting the primary link and watching the floating static install.

STOP • DEPTH LIMIT

No policy-based routing, no VRF-aware static routes, no track/IP SLA object tracking.

☐ STUDIED ☐ ANKI ☐ PROVEN

3.4 Configure and verify single area OSPFv2 ★ HIGH YIELD

CLASS [CONFIG]

YIELD HIGHEST SINGLE TECHNOLOGY ON THE EXAM

JEREMY DAYS 24, 26, 27, 28

- Enable OSPF, set a process ID (locally significant — know that), and advertise networks with a wildcard mask. Also know the per-interface method (ip ospf <pid> area <area>) as an alternative.
- Router ID selection order: manually configured RID → highest loopback IP → highest active physical interface IP. Know that changing it requires a process clear/reload to take effect.
- Neighbour states in order: Down → Init → 2-Way → ExStart → Exchange → Loading → Full. Know what happens at each and where a stuck adjacency points (stuck in 2-Way on a broadcast segment is normal between two DROTHERs; stuck in ExStart usually means MTU mismatch).
- Requirements for adjacency, all of which must match: area ID, hello and dead intervals, subnet/mask on the segment, authentication, MTU, and the interfaces must not be passive. Given a broken lab, checking these five in order is the whole troubleshooting method.
- Network types: broadcast (Ethernet — elects DR/BDR, hello 10 / dead 40) vs point-to-point (no DR/BDR, hello 10 / dead 40). Know how to change the interface network type and why you'd force point-to-point on an Ethernet link between exactly two routers.
- DR/BDR election: highest OSPF interface priority wins; tie broken by highest router ID. Priority 0 means never DR/BDR. The election is non-preemptive — a new higher-priority router does not take over until the current DR fails. This non-preemption point is a guaranteed question.
- Passive interfaces: stop hellos on a LAN with no OSPF neighbours while still advertising the subnet. Know passive-interface default and the per-interface exception form.
- Cost/metric: cost = reference bandwidth (100 Mbps default) ÷ interface bandwidth. Know why the default reference bandwidth is a problem on gigabit links and how to change it. Also know how to set cost manually per interface.
- Default route injection with default-information originate.
- Verify: show ip ospf neighbor (states, DR/BDR roles), show ip ospf interface brief, show ip ospf interface <int> (network type, cost, timers, priority), show ip protocols, show ip route ospf.

STOP • DEPTH LIMIT

Hard stop at single area. Do not study: multi-area design, ABR/ASBR behaviour, LSA types 1–7, virtual links, stub/NSSA/ totally-stubby areas, OSPFv3, OSPF authentication configuration, or route summarisation between areas. Jeremy covers some of this for completeness — watch, don't lab. Also: RIP and EIGRP (Jeremy Day 25) are not on the v1.1 blueprint at all. Watch once for AD context, then move on.

☐ STUDIED ☐ ANKI ☐ PROVEN

3.5 Describe the purpose, functions, and concepts of first hop redundancy protocols

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 29

- The problem FHRPs solve: an endpoint has one default gateway, so a redundant router is useless without a shared virtual IP.
- Virtual IP and virtual MAC concept — the hosts point at the VIP and never know a failover happened.

- HSRP (Cisco proprietary, active/standby, virtual MAC 0000.0C07.ACxx for v1, priority 100 default, preemption off by default), VRRP (open standard, master/backup, 0000.5E00.01xx), GLBP (Cisco, provides actual load balancing across multiple forwarders, not just redundancy).
- Load-balancing across FHRP groups by making different routers active for different VLANs.
- Note: v1.1 makes this a describe objective, but Lab 4 in this plan still has you configure HSRP — because building it is the fastest way to actually understand it, and it's a real-job skill.

STOP • DEPTH LIMIT

Describe verb only. No HSRP authentication, no interface tracking decrement tuning depth, no GLBP configuration.

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 4.0 — IP Services (10%)

Nine objectives for 10% of the exam — so each one is shallow. Resist depth here. Only 4.1, 4.2, 4.6, and 4.8 carry a configure verb.

4.1 Configure and verify inside source NAT using static and pools

CLASS [CONFIG]

YIELD MEDIUM-HIGH

JEREMY DAYS 44, 45

- The four address types and what each means: inside local, inside global, outside local, outside global. Fill in a NAT translation table from a scenario — this is the classic question.
- Mark inside and outside interfaces correctly (ip nat inside / ip nat outside). Forgetting this is the number one reason NAT "doesn't work" in a lab.
- Static NAT: a permanent one-to-one mapping.
- Dynamic NAT with a pool: define the pool, define an ACL matching the inside source addresses, bind them.
- PAT: overload a pool or overload the outside interface address; understand that port numbers distinguish the sessions.
- Verify with show ip nat translations and show ip nat statistics; clear ip nat translation * to reset.

STOP • DEPTH LIMIT

No NAT64, no CGNAT, no NAT virtual interfaces, no port forwarding beyond basic static NAT with port.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.2 Configure and verify NTP operating in client and server mode

CLASS [CONFIG]

YIELD LOW-MEDIUM

JEREMY DAY 37

- Point a device at an NTP server; configure a device as an NTP master with a stratum.
- Stratum concept: stratum 0 reference clock, each hop adds one, stratum 16 = unsynchronised.
- Why it matters operationally: log correlation, certificate validity, Kerberos.
- Verify with show ntp status (synchronised/unsynchronised, stratum) and show ntp associations (the * marks the chosen peer).
- Set the timezone and know clock set for manual time.

STOP • DEPTH LIMIT

No NTP authentication, no symmetric active/passive or broadcast modes in depth.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.3 Explain the role of DHCP and DNS within the network

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAYS 38, 39

- DHCP DORA sequence — Discover (broadcast), Offer, Request (broadcast), Acknowledge. Know which are broadcast and which are unicast, and the ports (67 server / 68 client).
- What a DHCP scope hands out: address, mask, default gateway, DNS servers, lease time, domain name.
- DNS: resolving a name to an address, recursive vs authoritative at a conceptual level, A vs AAAA vs CNAME vs MX vs PTR records, UDP 53 (TCP 53 for zone transfers/large responses).
- The troubleshooting logic: "can ping by IP but not by name" → DNS. "Client has 169.254.x.x" → DHCP unreachable.

STOP · DEPTH LIMIT

No DNSSEC, no zone file syntax, no BIND administration.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.4 Explain the function of SNMP in network operations

CLASS [CONCEPT]

YIELD LOW

JEREMY DAY 40

- Manager / agent / MIB / OID relationship.
- Get, GetNext, GetBulk, Set, Trap, Inform. Know that a trap is unacknowledged and an inform is acknowledged — that single distinction is the most-asked SNMP fact.
- Versions: v1 and v2c use cleartext community strings; v3 adds authentication, integrity, and encryption. v3 is the answer to any security-flavoured SNMP question.
- Ports 161 (polling) and 162 (traps/informs).

STOP · DEPTH LIMIT

No SNMP configuration (no configure verb), no MIB tree walking, no OID memorisation.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.5 Describe the use of syslog features, including facilities and severity levels

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 41

- Severity levels 0–7, memorised with a mnemonic: 0 Emergency, 1 Alert, 2 Critical, 3 Error, 4 Warning, 5 Notification, 6 Informational, 7 Debugging. ("Every Awesome Cisco Engineer Will Need Ice cream Daily.")
- Lower number = more severe. Configuring a logging level of 4 captures levels 0 through 4 — this inclusive behaviour is the exam trap.
- Message anatomy: %FACILITY-SEVERITY-MNEMONIC: description. Be able to read %LINK-3-UPDOWN and state the severity.
- Logging destinations: console, monitor (VTY), buffered, and a remote syslog server (UDP 514).
- Facility concept at a naming level only.

STOP · DEPTH LIMIT

No syslog server software administration, no RFC 5424 structured-data format.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.6 Configure and verify DHCP client and relay

CLASS [CONFIG]

YIELD MEDIUM

JEREMY DAY 39

- Configure a router interface as a DHCP client (obtain its address via DHCP).
- Configure a DHCP relay agent (ip helper-address) on the SVI/subinterface facing the clients, pointing at the server. Understand precisely why it's needed: DHCP Discover is a broadcast and routers don't forward broadcasts. The relay converts it to a unicast and inserts the client subnet so the server picks the right scope.
- Configure a Cisco router as a DHCP server: pool, network, default-router, dns-server, and excluded addresses. Know that you exclude before defining the pool in practice.
- Verify with show ip dhcp binding, show ip dhcp pool, show ip dhcp conflict, and confirm from the client side.

STOP · DEPTH LIMIT

No DHCP option 82 configuration depth (it belongs to 5.7 DHCP snooping awareness), no DHCPv6.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.7 Explain the forwarding per-hop behavior (PHB) for QoS

CLASS [CONCEPT] YIELD LOW-MEDIUM JEREMY DAYS 46, 47

- Classification (identify the traffic) and marking (set a value so downstream devices don't have to re-classify). CoS is the Layer 2 marking in the 802.1Q tag (3 bits, 0–7); DSCP is the Layer 3 marking in the IP ToS byte (6 bits).
- Key DSCP values: EF (Expedited Forwarding, DSCP 46) for voice; AF classes; CS classes; default 0.
- Queuing and congestion management: what a scheduler does, priority/LLQ for voice.
- Congestion avoidance: tail drop vs WRED.
- Policing vs shaping — policing drops or re-marks excess traffic immediately, shaping buffers it and sends it later. Policing is bursty, shaping is smooth. This comparison is the most likely QoS question.
- Trust boundary: why you trust an IP phone's markings but not a PC's.

STOP • DEPTH LIMIT

Explain verb only. No MQC configuration (class-map/policy-map/service-policy), no queue-depth tuning, no bandwidth-reservation maths.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.8 Configure network devices for remote access using SSH ★ HIGH YIELD

CLASS [CONFIG] YIELD HIGH JEREMY DAY 42

- The full, ordered sequence — you will be asked to produce it and a missing step breaks it: set a hostname, set a domain name, generate an RSA key pair (1024 bits or more for SSHv2), create a local user with a privilege level and secret, configure the VTY lines for local login and to accept SSH transport only, and force version 2.
- Why the hostname and domain name must be set first (they seed the key's fully qualified name).
- VTY line hardening: exec-timeout, login local, transport input ssh, and optionally an ACL applied with access-class.
- enable secret vs enable password, and service password-encryption (weak, type 7, reversible — know that it is not real security).
- Verify with show ip ssh, show ssh, show users, and by actually SSHing in from another device.

STOP • DEPTH LIMIT

No SSH public-key authentication, no certificate-based auth, no AAA server integration configuration.

☐ STUDIED ☐ ANKI ☐ PROVEN

4.9 Describe the capabilities and functions of TFTP/FTP in the network

CLASS [CONCEPT] YIELD LOW JEREMY DAY 43

- TFTP: UDP 69, no authentication, no directory listing, tiny and simple. Used for IOS image transfer, config backup, and PXE boot.
- FTP: TCP 20 (data) and 21 (control), authenticated, supports directory operations, active vs passive mode at a conceptual level.
- The network-engineering use case: backing up and restoring running/startup configs and IOS images to/from a server.

STOP • DEPTH LIMIT

Lowest-yield objective in the blueprint. One Anki pass, then leave it. No SFTP/FTPS/SCP depth, no copy command drilling beyond basic awareness.

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 5.0 — Security Fundamentals (15%)

Mostly conceptual, with three sharp exceptions: ACLs (5.6) and Layer 2 security (5.7) are heavily simulated, and 5.10 is the one remaining GUI configure task.

5.1 Define key security concepts (threats, vulnerabilities, exploits, mitigation)

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 48

- The four definitions and how they relate: a vulnerability is the weakness, a threat is the potential danger, an exploit is the mechanism used against the vulnerability, and mitigation is the countermeasure. Getting these four straight answers most questions here.
- CIA triad: confidentiality, integrity, availability — and which one a given attack breaks.
- Attack vocabulary: DoS/DDoS, reflection/amplification, spoofing (MAC, IP, DHCP, ARP), on-path/MITM, reconnaissance, social engineering (phishing, spear phishing, whaling, vishing, smishing), tailgating, password attacks (brute force, dictionary, credential stuffing), malware types.

STOP · DEPTH LIMIT

No threat-intelligence frameworks, no MITRE ATT&CK, no incident-response process models.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.2 Describe security program elements (user awareness, training, physical access control)

CLASS [CONCEPT]

YIELD LOW

JEREMY DAY 48

- User awareness (broad, passive — posters, simulated phishing) vs training (formal, structured, role-specific) vs physical access control (locks, badges, mantraps, cameras, secured racks).
- Distinguish these three cleanly. The questions here are almost always "which category does this example belong to."

STOP · DEPTH LIMIT

One Anki pass. No compliance framework study (ISO 27001, NIST CSF).

☐ STUDIED ☐ ANKI ☐ PROVEN

5.3 Configure and verify device access control using local passwords

CLASS [CONFIG]

YIELD MEDIUM

JEREMY DAYS 4, 48

- Console line password, VTY line password, enable secret vs enable password, local username/secret pairs with privilege levels.
- service password-encryption (type 7, reversible, cosmetic) vs secret (type 5 MD5 / type 8/9 scrypt — hashed, not reversible). Know which is actually secure.
- login vs login local and what each implies about where credentials come from.
- exec-timeout and logging synchronous on lines.
- Verify with show running-config | section line, and show privilege / show users.

STOP · DEPTH LIMIT

No AAA configuration (aaa new-model and TACACS+/RADIUS setup is beyond the configure scope — 2.8 and 5.8 only require describing/comparing them).

☐ STUDIED ☐ ANKI ☐ PROVEN

5.4 Describe security password policy elements

CLASS [CONCEPT]

YIELD LOW

JEREMY DAY 48

- Complexity, length, expiration/rotation, reuse history, lockout thresholds, secure storage.
- Multifactor authentication as the headline alternative — the three factors: something you know, something you have, something you are. Be able to identify whether a given pair is genuinely multifactor (two passwords is not).
- Certificates and biometrics as password alternatives, and the trade-offs of each (biometrics can't be changed once compromised).
- Password managers and passphrases.

STOP · DEPTH LIMIT

No PKI hierarchy depth, no certificate enrolment process, no biometric FAR/FRR maths.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.5 Describe IPsec remote access and site-to-site VPNs

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY V1.1 VPN VIDEO (WAN ARCHITECTURES / DAY 53 AREA)

- Site-to-site: gateway-to-gateway tunnel, always on, transparent to hosts, connects two whole networks. Remote-access: client-to-gateway, on-demand, per-user, typically SSL/TLS-based in modern deployments.
- What IPsec provides: confidentiality (encryption), integrity (hashing), authentication (pre-shared key or certificates), anti-replay.
- The two protocols: AH (authentication and integrity, no encryption) and ESP (encryption and integrity). ESP is what's actually used. That AH doesn't encrypt is the classic question.
- Tunnel mode (encrypts the whole original packet and adds a new IP header — used site-to-site) vs transport mode (encrypts payload only).
- IKE's role in negotiating the security association, at a one-paragraph level. GRE over IPsec: why GRE is added (it carries multicast/routing protocols, which plain IPsec can't).

STOP · DEPTH LIMIT

Describe verb only. No crypto map or IKEv2 proposal configuration. No DMVPN, no FlexVPN, no cipher-suite tables.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.6 Configure and verify access control lists ★ HIGH YIELD

CLASS [CONFIG]

YIELD VERY HIGH

JEREMY DAYS 34, 35

- Wildcard masks, fluently and in both directions. This is subnetting again, inverted. 0 = must match, 1 = don't care. Know the shortcuts: host = 0.0.0.0, any = 255.255.255.255.
- Standard ACLs (1–99, 1300–1999): match source only. Placement rule: as close to the destination as possible, because they'd otherwise block too much.
- Extended ACLs (100–199, 2000–2699): match source, destination, protocol, and port. Placement rule: as close to the source as possible, to drop unwanted traffic early.
- Named ACLs (standard and extended) — and the key operational advantage: sequence numbers let you insert or delete an individual line without rebuilding the list. Know how to do that.

- Implicit deny any at the end of every ACL. Every ACL needs at least one permit or it blocks everything. This causes more lab failures than anything else.
- Top-down first-match processing: order matters, and a more specific rule must come before a broader one.
- Applying an ACL: to an interface with a direction (in or out), and to VTY lines with access-class.
- Verify with show access-lists (read the match counters — they prove traffic is hitting the rules), show ip interface <int> (shows which ACL is applied where), and show running-config.

STOP • DEPTH LIMIT

No IPv6 ACLs, no time-based ACLs, no reflexive or dynamic (lock-and-key) ACLs, no object groups.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.7 Configure and verify Layer 2 security features (DHCP snooping, DAI, port security) ★ HIGH YIELD

CLASS [CONFIG]

YIELD HIGH

JEREMY DAYS 49, 50, 51

- Port security: enable it (requires the port be statically access or trunk first — it won't apply to a dynamic port), set maximum MAC count, choose the learning method (static, dynamic, or sticky), and set the violation mode.
- The three violation modes and their exact differences: protect (drops offending traffic silently, no log, no counter), restrict (drops, logs, increments the violation counter), shutdown (err-disables the port — this is the default). Know how to recover an err-disabled port (shut/no shut, or errdisable recovery).
- Aging and sticky MAC persistence (sticky entries go into the running-config, so they're lost on reload unless saved).
- DHCP snooping: the point is to stop a rogue DHCP server. Enable globally and per-VLAN, then designate trusted ports (uplinks toward the legitimate server and other switches); everything else is untrusted by default and has its DHCP server messages dropped. Know about rate limiting and Option 82 insertion, and the binding table it builds (MAC ↔ IP ↔ VLAN ↔ port ↔ lease).
- Dynamic ARP Inspection: stops ARP spoofing / on-path attacks. Enable per-VLAN, set trusted ports, and understand that DAI depends on the DHCP snooping binding table — which is why snooping must be configured first. Static ARP ACLs cover non-DHCP hosts.
- Verify: show port-security, show port-security interface <int>, show port-security address, show ip dhcp snooping, show ip dhcp snooping binding, show ip arp inspection.

STOP • DEPTH LIMIT

No 802.1X port-based authentication configuration, no MACsec, no storm control.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.8 Compare authentication, authorization, and accounting concepts

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 48

- Authentication = who are you. Authorization = what are you allowed to do. Accounting = what did you do (logging, auditing, billing).
- Map each to a scenario. Nearly all questions here are "which of the three is this an example of?"
- TACACS+ vs RADIUS comparison again (see 2.8) — the separation of AAA functions in TACACS+ is what makes it right for device administration.

STOP • DEPTH LIMIT

No ISE, no 802.1X supplicant/authenticator/authentication-server flow in depth.

☐ STUDIED ☐ ANKI ☐ PROVEN

5.9 Describe wireless security protocols (WPA, WPA2, WPA3)

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 57

- The progression and the cipher for each: WEP (broken, RC4), WPA (TKIP), WPA2 (AES-CCMP), WPA3 (AES-GCMP with SAE replacing the PSK handshake).
- Personal (PSK — shared secret, everyone uses the same one) vs Enterprise (802.1X + RADIUS — per-user credentials) for each generation.
- What WPA3 actually fixes: SAE (Dragonfly) defeats offline dictionary attacks against a captured handshake, and adds forward secrecy. Also know Enhanced Open (OWE) for open networks.
- The 802.1X roles: supplicant (client), authenticator (AP/WLC), authentication server (RADIUS).

STOP • DEPTH LIMIT

No four-way-handshake cryptographic breakdown, no EAP method comparison table (PEAP vs EAP-TLS vs EAP-FAST detail).

☐ STUDIED ☐ ANKI ☐ PROVEN

5.10 Configure and verify WLAN within the GUI using WPA2 PSK

CLASS [CONFIG] (GUI)

YIELD MEDIUM

JEREMY DAY 58

Master

- The complete end-to-end flow in a WLC GUI: create the dynamic interface and map it to a VLAN → create the WLAN with SSID and profile name → set the interface on the General tab → on Security → Layer 2, choose WPA2 with AES and PSK → enter the pre-shared key → enable the WLAN → verify a client associates.
- Practise this in Packet Tracer against the WLC device. The GUI there is simplified but the flow matches.
- Verify: client shows connected, receives an address from the correct VLAN's scope, and can ping its gateway.

STOP • DEPTH LIMIT

No WPA2-Enterprise/RADIUS setup, no per-firmware-version menu-path memorisation. WPA2 PSK only — that's literally what the objective says.

☐ STUDIED ☐ ANKI ☐ PROVEN

Domain 6.0 — Automation And Programmability (10%)

Zero configure verbs. Every objective is explain, compare, describe, or recognize. This is pure reading and flashcards — and it is the highest points-per-hour domain on the entire exam, because 10% of the grade requires no lab time at all. Do not skip it, and do not over-engineer it.

6.1 Explain how automation impacts network management

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 59

- Traditional per-device CLI management: slow, inconsistent, error-prone, doesn't scale, configuration drift.
- What automation buys: consistency, speed, scale, version control, repeatability, fewer human errors.
- What it costs: upfront skill investment, tooling complexity, and the fact that a mistake now propagates to every device instantly.

STOP • DEPTH LIMIT

No actual scripting. No Python required for CCNA.

☐ STUDIED ☐ ANKI ☐ PROVEN

6.2 Compare traditional networks with controller-based networking

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 62

- Traditional: every device runs its own control plane and data plane; configuration is per-box.
- Controller-based: the control plane is centralised in a controller which has a full network view and programs the devices; the devices keep their data planes.
- Intent-based networking: you declare the outcome, the controller works out the device configuration and continuously verifies it.

STOP • DEPTH LIMIT

No Catalyst Center / DNA Center hands-on, no SD-Access fabric roles in depth.

☐ STUDIED ☐ ANKI ☐ PROVEN

6.3 Describe controller-based, software defined architecture (overlay, underlay, fabric)

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 62

Master

- Underlay = the physical network and the routing that makes it reachable. Overlay = the virtual/tunnelled network built on top (VXLAN, GRE). Fabric = underlay + overlay together as one managed system.
- Control plane vs data plane separation: control plane decides where traffic goes (routing protocols, STP, ARP), data plane actually moves the frames. SDN lifts the control plane out to the controller. Also know the management plane (SSH, SNMP, syslog).
- Northbound API: controller ↔ applications/operators, typically REST/JSON, human-and-app friendly. Southbound API: controller ↔ network devices, typically NETCONF/RESTCONF/OpenFlow. Which direction is which is a guaranteed question — northbound points up toward the user.

STOP • DEPTH LIMIT

No VXLAN header structure, no OpenFlow table pipeline, no LISP.

6.4 Explain AI (generative and predictive) and machine learning in network operations

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY V1.1 AI/ML VIDEO (NEW IN V1.1)

- Predictive AI forecasts from patterns in existing data (capacity planning, anomaly detection, predicting a failing link). Generative AI produces new content (config drafts, summaries, natural-language queries against network data).
- Supervised (labelled data) vs unsupervised (finds structure itself, good for anomaly detection) vs reinforcement learning.
- AIOps in practice: baselining normal behaviour, flagging deviation, correlating events into a single root cause, recommending remediation.
- Limitations and risks: training data quality, hallucination in generative output, and why a human stays in the loop for network changes.

STOP • DEPTH LIMIT

New v1.1 objective, low depth expected. No model architectures, no training methodology, no vendor product feature lists.

6.5 Describe characteristics of REST-based APIs

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 61

- CRUD ↔ HTTP verb mapping: Create = POST, Read = GET, Update = PUT/PATCH, Delete = DELETE. Know PUT (replace whole resource) vs PATCH (partial update).
- REST constraints: client-server, stateless (each request carries everything it needs — this is the most-tested property), cacheable, uniform interface, layered.
- HTTP status code families: 2xx success (200 OK, 201 Created), 3xx redirect, 4xx client error (400 bad request, 401 unauthorised, 403 forbidden, 404 not found), 5xx server error.
- Authentication types: Basic (base64 user:pass — not encryption), Bearer token, API key, OAuth 2.0.
- Data encoding: JSON (most common), XML, YAML. Request/response headers and body.
- URI anatomy and what a resource path represents.

STOP • DEPTH LIMIT

No writing API client code, no NETCONF YANG model structure, no gRPC.

6.6 Recognize the capabilities of configuration management mechanisms such as Ansible and Terraform

CLASS [CONCEPT]

YIELD LOW-MEDIUM

JEREMY DAY 63 + V1.1 TERRAFORM VIDEO

- Ansible: agentless, push model, uses SSH, written in YAML (playbooks, inventory, modules), procedural/imperative-leaning, focused on configuration management of existing infrastructure.
- Terraform: agentless, uses provider APIs, written in HCL, declarative, maintains state, focused on provisioning infrastructure.
- The comparison axes the exam cares about: agent vs agentless, push vs pull, provisioning vs configuration management, declarative vs procedural, mutable vs immutable infrastructure, and the language/file format each uses.
- Puppet and Chef for contrast: agent-based, pull model, Ruby-based DSL.

STOP · DEPTH LIMIT

Recognize verb — the lowest bar in the blueprint. No playbook authoring, no HCL syntax, no module reference.

☐ STUDIED ☐ ANKI ☐ PROVEN

6.7 Recognize components of JSON-encoded data

CLASS [CONCEPT]

YIELD MEDIUM

JEREMY DAY 60

- The six data types: string, number, boolean, null, object {}, array [].
- Syntax rules: keys are always double-quoted strings, key-value pairs separated by colons, elements by commas, no trailing comma, no comments.
- The exam skill: given a block of JSON, answer "what is the value of key X" or "how many elements are in array Y" or "which of these is syntactically valid JSON." Practise reading nested structures — an array of objects inside an object is the standard shape.
- Recognise XML and YAML on sight and be able to tell them apart from JSON.

STOP · DEPTH LIMIT

Recognize verb. No parsing code, no JSON Schema, no YAML anchors/aliases.

☐ STUDIED ☐ ANKI ☐ PROVEN

High-Yield Summary — Where The Points Actually Are

If time collapses, protect these in this order. They span roughly two-thirds of the scoring weight and virtually all of the simulation items.

RANK	TOPIC	OBJECTIVES	WHY
1	IPv4 subnetting & VLSM	1.6	Underpins ACLs, OSPF wildcards, NAT pools, DHCP scopes, routing tables. A subnetting error cascades into every other domain.
2	Single-area OSPFv2	3.4	The densest configure-and-verify technology, in the largest domain.
3	VLANs & 802.1Q trunking	2.1, 2.2	The foundation of nearly every switching simulation.
4	ACLs (standard, extended, named)	5.6	Heavily simulated, and unforgiving — wildcard mask or placement errors fail the whole item.
5	Routing table interpretation & forwarding logic	3.1, 3.2	Longest prefix match + AD + metric appears in some form on nearly every attempt.
6	Rapid PVST+ interpretation	2.5	Root election and port roles from a diagram, by hand.
7	IPv4/IPv6 static & floating static routing	3.3	Fast points, easy to lab, easy to get wrong under time pressure.
8	EtherChannel (LACP)	2.4	Mode matrix + consistency troubleshooting.

RANK	TOPIC	OBJECTIVES	WHY
9	IPv6 addressing & types	1.8, 1.9	Compression and EUI-64 are pure speed skills.
10	Layer 2 security (port security, DHCP snooping, DAI)	5.7	Configure-verb security objectives with clear, testable behaviour.
11	SSH device hardening	4.8	A fixed command sequence that appears constantly.
12	NAT/PAT	4.1	The inside-local/inside-global table question is near-guaranteed.
13	Domain 6.0 in full	6.1–6.7	10% of the exam for zero lab time. The best points-per-hour on the blueprint.

MASTER "DO NOT STUDY" LIST Everything below is out of scope for 200-301 v1.1. Some of it appears in Jeremy's course or the OCG for context. Watch once if you like; never lab it, never flashcard it.

- ☐ RIP and EIGRP configuration (not on the v1.1 blueprint — Jeremy Day 25 is context only)
- ☐ BGP in any form
- ☐ OSPF multi-area, LSA types, ABR/ASBR, stub/NSSA areas, virtual links, OSPF authentication
- ☐ OSPFv3 / IPv6 routing protocols
- ☐ VTP configuration (Day 19 is context only)
- ☐ MST, legacy PVST+ timer tuning
- ☐ IPv6 ACLs, time-based ACLs, reflexive ACLs, object groups
- ☐ DHCPv6, SLAAC internals, NAT64, tunnelling mechanisms
- ☐ MQC QoS configuration (class-map / policy-map / service-policy)
- ☐ AAA server configuration (aaa new-model, TACACS+/RADIUS setup)
- ☐ 802.1X configuration, MACsec, storm control
- ☐ IPsec crypto map / IKEv2 configuration, DMVPN
- ☐ Policy-based routing, VRF configuration, IP SLA / object tracking
- ☐ VXLAN, LISP, SD-Access fabric roles, ACI

- ☐ Kubernetes, Docker commands, ESXi administration
 - ☐ Python scripting, NETCONF/YANG models, Ansible playbook authoring
 - ☐ Wireless: 802.11 frame types, RF maths, site surveys, EAP method comparison tables
 - ☐ Any Cisco hardware model numbers or IOS version feature matrices
-